Chapter test

10 Nutrition and gas exchange in plants

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

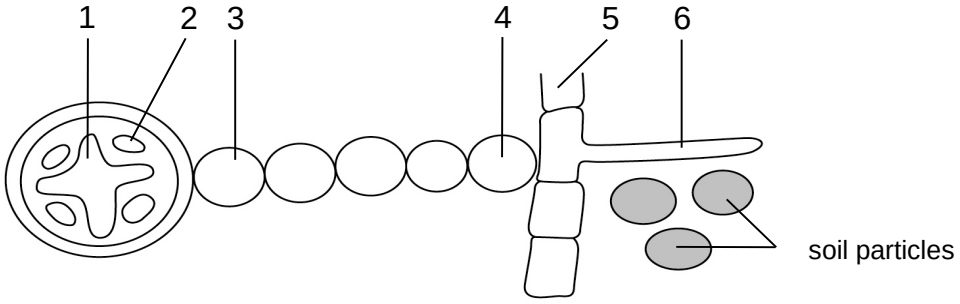
Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. What would happen if the plants lack magnesium?

   A. Black patches appear on leaves.
   B. Young leaves become long and thin.
   C. Leaves become yellow.
   D. Leaves become dark green. ☐

***Directions*:** Questions 2 to 4 refer to the diagram below, which shows the transverse section of a dicotyledonous root.



★ 2. Which of the following is/are the function(s) of cell 4?

   (1) store food
   (2) carry out photosynthesis
   (3) allow water and minerals to pass through

   A. (3) only
   B. (1) and (2) only
   C. (1) and (3) only
   D. (2) and (3) only ☐

© 2019 Aristo Educational Press Ltd.

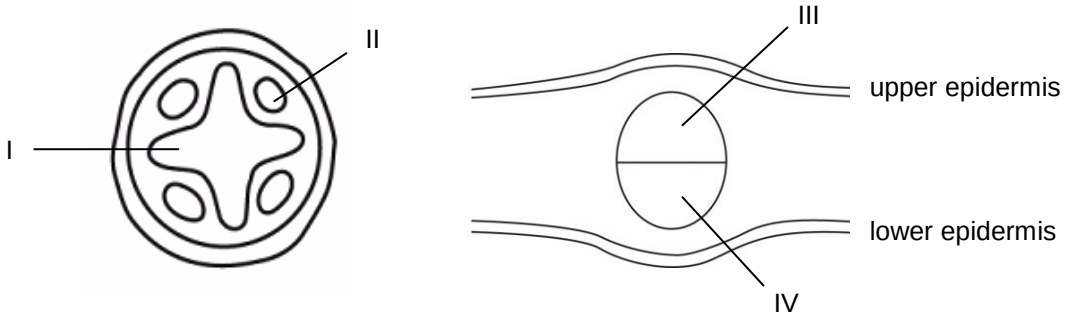
★ 3. By which of the following processes are minerals absorbed into cell 6?

A. transpiration
B. diffusion
C. osmosis
D. active transport ☐

★★ 4. Which of the following cells has the lowest water potential when the root is absorbing water from the soil?

A. 3
B. 4
C. 5
D. 6 ☐

5. The diagrams below show the sections of the root and the leaf of a dicotyledonous plant.



Root **Leaf**

Which of the labelled tissues are xylem?

A. I and III
B. I and IV
C. II and III
D. II and IV ☐

★ 6. By which of the following processes does oxygen move into the cytoplasm of a root hair cell?

A. transpiration
B. diffusion
C. osmosis
D. active transport ☐

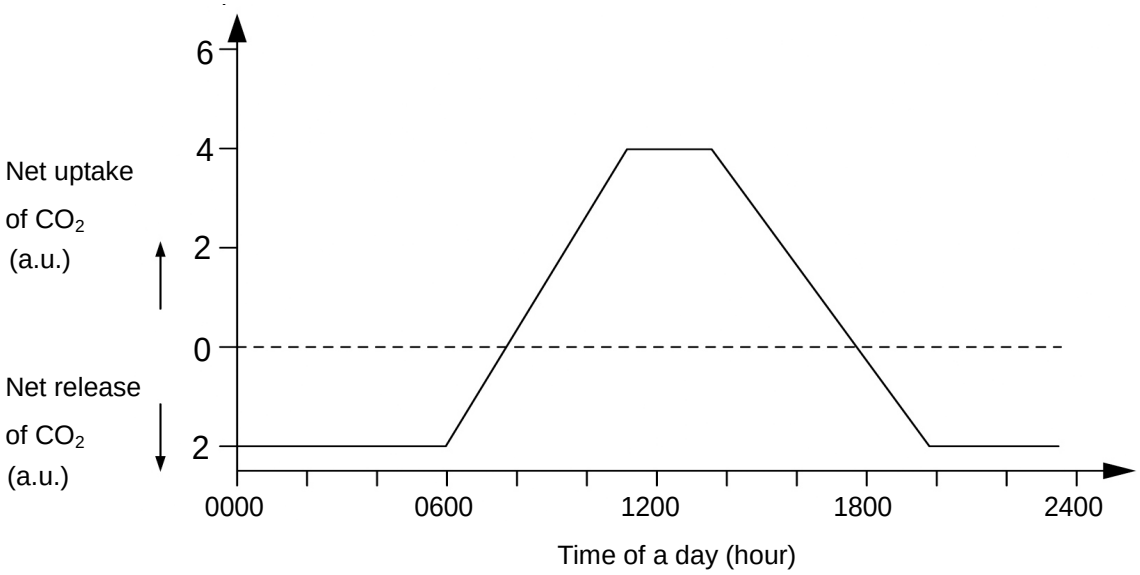
© 2019 Aristo Educational Press Ltd.

★ 7. Which types of cells in the leaf of a terrestrial plant can carry out photosynthesis?

(1) spongy mesophyll cells
(2) guard cells
(3) root hair cells

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

***Directions*:** Questions 8 to 10 refer to the graph below, which shows the net uptake and net release of carbon dioxide of a dicotyledonous plant in a day.



★ 8. The plant carried out photosynthesis

A. from 0600 to 1800.
B. from 0600 to 2000.
C. from 0800 to 1800.
D. from 0800 to 2000.

★ 9. The plant carried out respiration

A. from 2400 to 0600 and from 2000 to 2400.
B. from 2400 to 0800 and from 1800 to 2400.
C. from 1100 to 1300.
D. all the time.

© 2019 Aristo Educational Press Ltd.

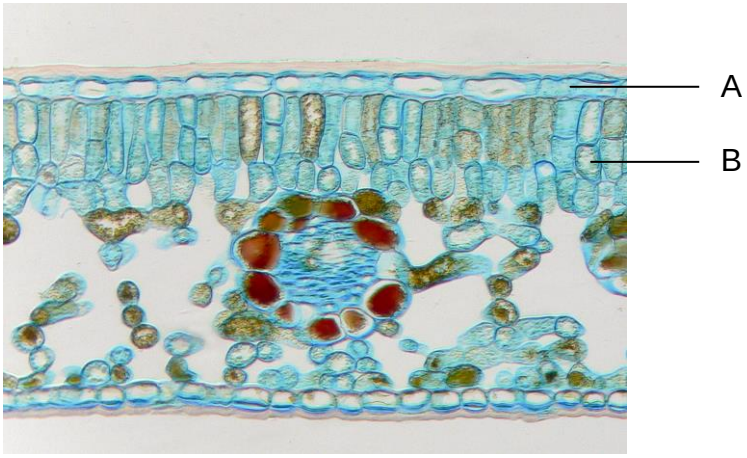
★★10. The highest rate of photosynthesis of the plant is

A. 2 a.u.
B. 4 a.u.
C. 6 a.u.
D. 8 a.u. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	10.1	10.1	10.1	10.1	10.2	10.2	10.2	10.2	10.2	10.2

B. Problem solving (19 marks)

★11. The photomicrograph below shows the section of a leaf.



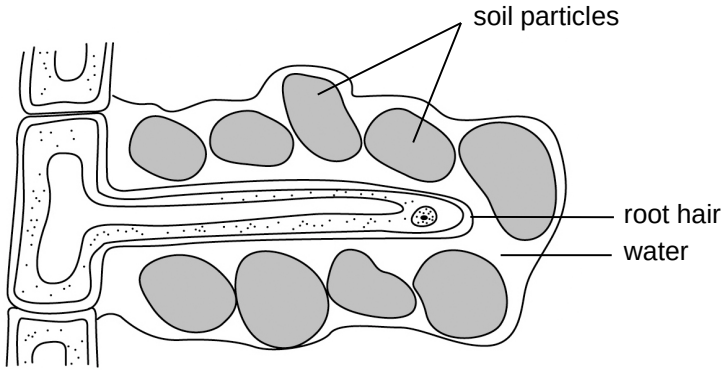
(a) Name cells A and B. State ***one*** difference in structure between cell A and cell B. (3 marks)

(b) With reference to the photomicrograph, describe ***one*** observable feature of the leaf and explain how this facilitates gas exchange between the leaf and the environment. (2 marks)

(Total: 5 marks)

© 2019 Aristo Educational Press Ltd.

★12. The diagram below shows a sectional view of a root hair and the surrounding soil particles.



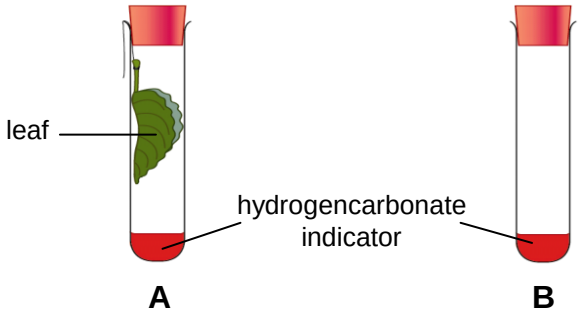
(a) With reference to the diagram, state ***two*** adaptations of the root hairs for water absorption. (4 marks)

(b) Describe how soil water passes into the root hair. (3 marks)

(Total: 7 marks)

© 2019 Aristo Educational Press Ltd.

★ 13. Mary carried out an experiment to investigate the light compensation point of a leaf. The diagram below shows the set-ups of the experiment.



Mary carried out the experiment under different light intensities. At each of the light intensity, two new leaves are placed into the two test tubes, A and B, and the two tubes were illuminated for 30 minutes. At the beginning of the experiment, the hydrogencarbonate indicator was red in colour in both test tubes.

(a) State ***two*** variables that Mary should keep constant throughout the experiment. (2 marks)

(b) What is the purpose of setting up test tube B? (1 mark)

The table below shows the results of the experiment.

	Light intensity (arbitrary units)				
	0	5	10	15	20
Colour of the hydrogencarbonate indicator in tube A	yellow	yellowish orange	orange	red	purple
Colour of the hydrogencarbonate indicator in tube B	red	red	red	red	red

(c) With reference to the results of the experiment, what is the light compensation point of the plant? Explain your answer. (3 marks)

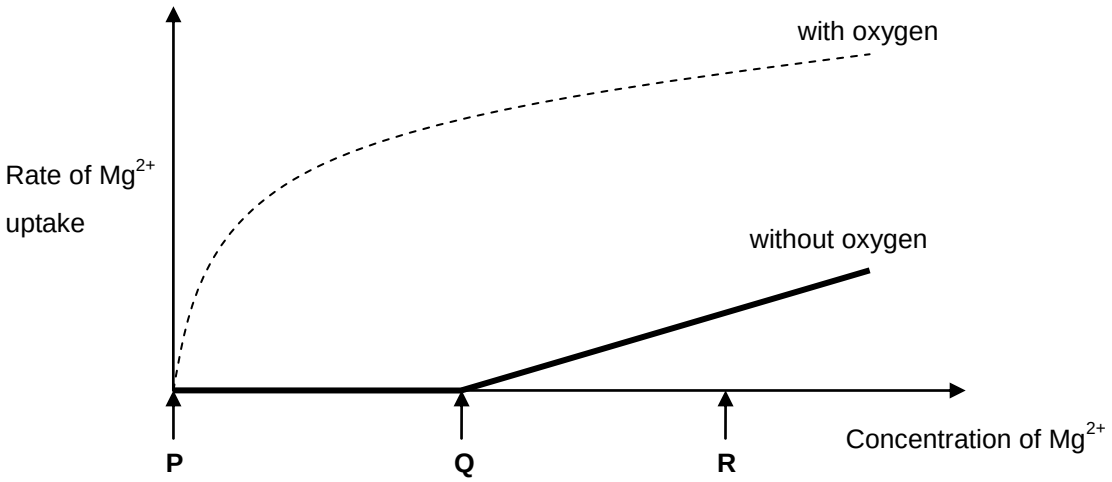
© 2019 Aristo Educational Press Ltd.

(d) Limewater is also an indicator for carbon dioxide. Explain why it is not suitable for use in this experiment. (1 mark)

(Total: 7 marks)

C. Exam challenge (11 marks)

★★14. The graph below shows the rate of magnesium ion (Mg^{2+}) uptake by a plant soaking in Mg^{2+} solution of different concentrations.



(a) As the concentration of the Mg^{2+} solution lies between P and Q, by what process does the plant absorb magnesium ions in the presence of oxygen? Explain your answer. (3 marks)

(b) Describe and explain the rate of magnesium ion uptake of the plant in the absence of oxygen as the concentration of the Mg^{2+} lies between

(i) P and Q; (3 marks)

© 2019 Aristo Educational Press Ltd.

(ii) Q and R. (3 marks)

(c) Plants obtain minerals from the soil. Soil contains many types of minerals. Are all minerals absorbed at the same rate or at different rates? Explain your answer with your biological knowledge. (2 marks)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	C	3.	D	4.	A	5.	A
6.	B	7.	B	8.	B	9.	D	10.	C

(1 mark each)

B. Problem solving (19 marks)

11. (a) Cell A: epidermal cell (1)
Cell B: palisade mesophyll cell (1)
Cell B contains chloroplasts while cell A does not. (1)

(b) The cells in the spongy mesophyll are loosely packed. / There are many air spaces in the spongy mesophyll. (1)
This allows gases to circulate freely inside the leaf. (1)

(Total: 5 marks)

12. (a)

Adaptation	Reason
• It is long and fine. (1)	• increase the surface area for water absorption / allow root to penetrate the spaces between soil particles (1)
• It is not covered by cuticle. (1)	• water can be absorbed easily (1)

(b) The cell sap and cytoplasm of the root hair usually contain dissolved substances. (1)
Hence, the water potential of soil water is higher than that of the root hair. (1)
Water enters the root hair by osmosis. (1)

(Total: 7 marks)

13. (a) Temperature (1)
Volume of hydrogencarbonate indicator added (1)

(b) It serves as control. (1)

(c) 15 arbitrary units (1)
There is no change in the colour of the hydrogencarbonate indicator in tube A at light intensity of 15 arbitrary units. (1)
This indicates that at this light intensity, there is no net uptake or release of carbon dioxide in the leaf. The rates of respiration and photosynthesis are equal. (1)

© 2019 Aristo Educational Press Ltd.

(d) Limewater can only show an increase in carbon dioxide content but cannot show a decrease. / Limewater cannot show the amount of carbon dioxide quantitatively. / Limewater can only show whether carbon dioxide is present or not. (1)

(Total: 7 marks)

C. Exam challenge (11 marks)

14. (a) By active transport (1)
The absorption of Mg^{2+} occurs only in the presence of oxygen, (1)
which suggests that the process of absorption requires energy from respiration. (1)

(b) (i) There is no uptake of Mg^{2+} between concentrations P and Q. (1)
Mg^{2+} concentration inside the cells is higher than that of the Mg^{2+} solution. (1)
In the absence of oxygen, active transport cannot occur and the cells cannot absorb Mg^{2+} ions against the concentration gradient. (1)

(ii) The rate of Mg^{2+} uptake increases between concentrations Q and R. (1)
Mg^{2+} concentration inside the cells is lower than that of the Mg^{2+} solution. (1)
Mg^{2+} ions diffuse into the plant cells along the concentration gradient. (1)

(c) Minerals are absorbed at different rates. (1)
Minerals move across the cell membrane through specific membrane proteins and the abundance of various membrane proteins may be different. (1)
(Or other acceptable answers)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.